PRODUCT REQUIREMENTS DOCUMENT

Mikrotik Hotspot Voucher Manager

Mobile Application

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1. Executive Summary

1.1 Problem Statement

Internet cafe owners and small hotspot business operators who rely on Mikrotik routers for their Wi-Fi hotspot services currently face a significant operational bottleneck: the creation and management of hotspot vouchers requires physical access to the router or a visit to a network technician. This process is time-consuming, costly, and inefficient, especially for operators managing multiple locations or those in remote areas far from technical support.

Each time an operator runs out of vouchers or needs to adjust session parameters, business operations are disrupted. Customers wait, revenue is lost, and the operator incurs travel costs. There is no affordable, mobile-first solution that allows these operators to remotely manage their Mikrotik routers and voucher systems.

1.2 Proposed Solution

The Mikrotik Hotspot Voucher Manager is a mobile application for iOS and Android that empowers operators to remotely manage their Mikrotik routers and create hotspot vouchers directly from their smartphones. The app communicates with a centralized VPS backend that runs FreeRADIUS for voucher authentication and accounting, and maintains secure WireGuard VPN tunnels to each registered Mikrotik router. Vouchers are created and stored centrally in the RADIUS database rather than as local users on each router, enabling consistent management across multiple locations without exposing router credentials or requiring complex network configurations on the operator's end.

1.3 Target Users

- Internet cafe owners and small hotspot business operators who sell Wi-Fi access via voucher codes
- Network administrators managing multiple Mikrotik router deployments across different sites
- Technicians and service providers who support multiple client routers and need centralized management

1.4 Business Objectives

- Eliminate the need for physical technician visits by enabling full remote voucher management
- Create a recurring revenue stream through tiered subscription plans
- Reach 100 active paying subscribers within 3 months of launch
- Reduce average voucher creation time to under 1 minute from the mobile device
- Achieve 90% reduction in technician visits for voucher-related tasks

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2. System Architecture

2.1 High-Level Architecture

The system follows a client-server model with five primary components communicating through secure channels. The mobile app serves as the client interface, the VPS backend acts as the API gateway and business logic layer, FreeRADIUS handles all voucher authentication and accounting centrally, WireGuard VPN tunnels provide encrypted connectivity to customer routers, and the Mikrotik RouterOS API enables programmatic router management. Vouchers are stored and authenticated entirely through FreeRADIUS on the VPS — not as local Mikrotik hotspot users — enabling centralized control, consistent accounting, and multi-router management from a single database.

2.2 Component Breakdown

2.2.1 Mobile Application (Client)

The mobile app is the primary user interface. It will be built as a cross-platform application (React Native or Flutter) targeting both iOS and Android. The app communicates exclusively with the VPS backend via HTTPS REST API calls. It does not connect directly to any Mikrotik router.

Attribute	Detail
Framework	React Native or Flutter (cross-platform)
Minimum OS	Android 8.0 (API 26) / iOS 13.0
Device Requirements	2 GB RAM minimum, ARM64 processor
Network	HTTPS over cellular or Wi-Fi
Authentication	JWT tokens stored in secure storage (Keychain / EncryptedSharedPreferences)
State Management	Redux / Provider pattern with offline caching
Push Notifications	Firebase Cloud Messaging (FCM) for Android, APNs for iOS

2.2.2 VPS Backend (Server)

The backend is a RESTful API server hosted on a Virtual Private Server. It handles authentication, subscription management, request routing to the correct router, and data persistence. It hosts the FreeRADIUS server that acts as the centralized authentication and accounting engine for all hotspot vouchers. It also manages WireGuard tunnel configurations and monitors router connectivity status.

Attribute	Detail

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Language / Framework	Node.js with Express.js or Python with FastAPI
Database	PostgreSQL for relational data and RADIUS backend (via rlm_sql), Redis for caching and session tokens
RADIUS Server	FreeRADIUS 3.x with SQL module (rlm_sql_postgresql) for user/voucher storage and accounting
API Style	RESTful with JSON payloads, versioned (e.g., /api/v1/)
Authentication	JWT with refresh token rotation, bcrypt password hashing
Hosting	Ubuntu 22.04 LTS on VPS (DigitalOcean, Hetzner, or equivalent)
Monitoring	Prometheus metrics, Grafana dashboards, structured logging
Uptime Target	99.5% availability

2.2.3 WireGuard VPN Layer

Each Mikrotik router is connected to the VPS through a persistent WireGuard VPN tunnel. This ensures that routers behind NAT or dynamic IPs remain accessible from the backend without requiring port forwarding or public IP addresses on the customer's network.

Attribute	Detail
Protocol	WireGuard (kernel-level, high performance)
Encryption	Curve25519 key exchange, ChaCha20-Poly1305 encryption
Tunnel Management	Automated key pair generation and config distribution
Keepalive	25-second persistent keepalive to maintain NAT traversal
IP Addressing	Private /30 subnets per tunnel (e.g., 10.10.x.0/30)
Failover	Automatic reconnection with exponential backoff

2.2.4 RouterOS API Integration

The VPS backend communicates with each Mikrotik router using the RouterOS API protocol over the WireGuard tunnel. With the RADIUS-based architecture, the RouterOS API is used primarily for router configuration, session monitoring, and system health checks — not for voucher storage. Each Mikrotik hotspot is configured to authenticate users against the centralized FreeRADIUS server via the RADIUS protocol.

Attribute	Detail
Protocol	RouterOS API (TCP port 8728) or API-SSL (TCP port 8729)
Library	routeros-client (Node.js) or librouteros (Python)
Authentication	Per-router API credentials stored encrypted in database

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Command Scope	Hotspot RADIUS config, session monitoring, CoA/disconnect, system resource queries
Timeout	5-second connection timeout, 10-second command timeout
Security	API access restricted to WireGuard tunnel IP only

2.2.5 FreeRADIUS Server

FreeRADIUS is the centralized authentication, authorization, and accounting (AAA) server running on the VPS. All Mikrotik routers are configured as RADIUS clients, pointing their hotspot authentication to the FreeRADIUS server over the WireGuard tunnel. Vouchers are stored as RADIUS users in the PostgreSQL database, and all session accounting data (time used, data consumed, login/logout events) flows into the same database via RADIUS accounting packets.

Attribute	Detail
Software	FreeRADIUS 3.x (stable release)
Backend Module	rlm_sql with rlm_sql_postgresql driver
Auth Protocol	PAP/CHAP/MS-CHAPv2 (configurable per deployment)
Accounting	RADIUS accounting (Start, Interim-Update, Stop) stored in radacct table
User Storage	radcheck / radreply tables in PostgreSQL (standard FreeRADIUS SQL schema)
Group Profiles	radusergroup / radgroupcheck / radgroupreply for bandwidth and limit profiles
CoA / Disconnect	RADIUS Change of Authorization (RFC 5176) for live session disconnect and limit updates
Simultaneous-Use	Enforced via FreeRADIUS Simultaneous-Use attribute to control shared sessions
RADIUS Clients	Each Mikrotik router registered as a NAS client with shared secret over WireGuard IP
Ports	Authentication: UDP 1812, Accounting: UDP 1813, CoA: UDP 3799

2.3 Data Flow

The end-to-end flow for a voucher creation request follows this path:

1. Operator opens the mobile app and authenticates via JWT.
2. Operator selects a router and configures voucher parameters (time limit, data limit, quantity, RADIUS group profile).
3. App sends HTTPS POST request to the VPS backend API.

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4. Backend validates the JWT, checks subscription quota, and verifies the target router is registered.
5. Backend inserts the voucher credentials into the FreeRADIUS SQL database (radcheck, radreply, radusergroup tables).
6. Backend associates the voucher with the target router for access control and reporting.
7. Backend logs the operation, updates quota counters, and returns the result to the mobile app.
8. App displays the created voucher credentials and offers print/share options.
9. When a customer connects to the hotspot, Mikrotik sends a RADIUS Access-Request to FreeRADIUS for authentication.
10. FreeRADIUS validates credentials, applies group profile limits (bandwidth, time, data), and returns Access-Accept.
11. Mikrotik sends RADIUS Accounting packets (Start, Interim-Update, Stop) to track session usage in the radacct table.

2.4 Security Architecture

- All client-to-server communication encrypted via TLS 1.3 (HTTPS)
- All server-to-router communication encrypted via WireGuard VPN
- RADIUS shared secrets per router (unique per NAS client), transmitted only over WireGuard tunnel
- Router API credentials encrypted at rest using AES-256 in the database
- JWT access tokens expire after 15 minutes; refresh tokens after 7 days
- Password hashing using bcrypt with a minimum cost factor of 12
- RADIUS user passwords stored with NT-Password hash or Cleartext-Password (required for CHAP) in encrypted database columns
- Rate limiting on all API endpoints (100 requests/minute per user)
- Input validation and sanitization on all user-supplied data
- Router API access and RADIUS traffic restricted to the WireGuard tunnel interface only

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3. Detailed Feature Specifications

3.1 Authentication & User Management

3.1.1 User Registration

Field	Type	Validation
Full Name	Text	Required, 2–100 characters
Email Address	Email	Required, unique, RFC 5322 compliant
Phone Number	Phone	Required, E.164 format with country code
Password	Password	Required, minimum 8 characters, at least 1 uppercase, 1 number
Business Name	Text	Optional, 2–200 characters
Country / Region	Dropdown	Required, used for localization and payment context

Upon registration, the system sends a verification email with a 6-digit OTP code. The user must verify their email within 24 hours to activate the account. Unverified accounts are purged after 72 hours.

3.1.2 Login

- Email + password authentication with JWT issuance
- Support for biometric login (fingerprint / Face ID) as a convenience layer after initial authentication
- Account lockout after 5 consecutive failed attempts (15-minute cooldown)
- Session management: only one active session per device; new login invalidates previous session on same device

3.1.3 Password Reset

Password reset is initiated via the “Forgot Password” flow. The system sends a time-limited (15-minute) OTP to the registered email. The user enters the OTP and sets a new password. The reset invalidates all existing sessions.

3.2 Subscription Management

Subscriptions control access to core features and enforce resource limits. Payment is handled manually through local bank transfers to keep transaction costs low and support operators in regions without automated payment infrastructure.

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3.2.1 Subscription Tiers

Feature	Starter	Professional	Enterprise
Monthly Price	$5 / month	$12 / month	$25 / month
Max Routers	1	3	10
Monthly Voucher Quota	500	2,000	Unlimited
Session Monitoring	Basic (active only)	Full (active + history)	Full + export
Dashboard	Basic stats	Advanced analytics	Full analytics + reports
Support	Email only	Email + priority queue	Dedicated support channel
Subscription Duration	1 month	1 or 2 months	1, 2, or 6 months

3.2.2 Payment Flow

1. Operator selects a subscription tier and duration in the app.
2. App displays bank transfer details (account number, reference code, amount).
3. Operator completes the transfer via their local banking app.
4. Operator uploads a screenshot or receipt of the payment in the app.
5. System notifies the platform admin of the pending payment.
6. Admin verifies the payment and manually activates the subscription.
7. Operator receives a push notification confirming activation.

3.2.3 Subscription Lifecycle

- Active: Full access to all features within the tier limits
- Expiring: Warning notifications sent at 7, 3, and 1 day before expiry
- Expired: Read-only access for 7 days (can view data but cannot create vouchers)
- Suspended: Account locked after 7-day grace period; data retained for 90 days
- Cancelled: Operator-initiated; access continues until current period ends

3.3 Router Management

3.3.1 Adding a Router

When an operator adds a new Mikrotik router, the system generates a unique WireGuard configuration and RADIUS client configuration for that router. The operator must apply the WireGuard config on the Mikrotik router and configure the hotspot server to use RADIUS authentication pointing to the VPS. Once applied, the router establishes a VPN tunnel and can authenticate hotspot users via FreeRADIUS.

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Field	Description	Source
Router Name	User-defined label (e.g., “Main Cafe Router”)	Operator input
Router Model	Auto-detected after connection (e.g., RB750Gr3)	RouterOS API
RouterOS Version	Auto-detected after connection	RouterOS API
API Username	Dedicated API user on the router	Operator input
API Password	Encrypted and stored on the VPS	Operator input
WireGuard Public Key	Generated by the VPS for this router	System generated
WireGuard Config	Downloadable/copyable config for the router	System generated
Tunnel IP	Assigned private IP for the VPN tunnel	System generated
RADIUS Shared Secret	Unique secret for RADIUS client authentication	System generated
RADIUS Server IP	VPS-side WireGuard tunnel IP for RADIUS auth	System generated
NAS Identifier	Unique identifier for this router as a RADIUS NAS	System generated

The system provides a step-by-step setup guide within the app, including: WireGuard peer configuration commands, Hotspot RADIUS server configuration (IP, ports 1812/1813, shared secret), enabling RADIUS accounting with 300-second Interim-Update interval, and optionally configuring RADIUS CoA on port 3799 for remote disconnect capability.

3.3.2 Router Status Monitoring

- Online: WireGuard handshake within last 150 seconds; API responds to ping
- Offline: No WireGuard handshake for over 150 seconds
- Degraded: WireGuard connected but API unresponsive (possible high load or misconfiguration)

Status checks run every 60 seconds. If a router transitions from Online to Offline, a push notification is sent to the operator after a 3-minute grace period to avoid false alarms from brief connectivity hiccups.

3.3.3 Router Operations

- Edit router name and API credentials
- Regenerate WireGuard keys (invalidates previous tunnel; requires reconfiguration on the router)
- Remove router (deletes WireGuard tunnel, removes all associated data after confirmation)
- View router system info: uptime, CPU load, memory usage, firmware version

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3.4 Voucher & User Management (FreeRADIUS-Based)

This is the core feature of the application. Unlike traditional Mikrotik hotspot setups where vouchers are stored as local users on each router, this system stores all vouchers centrally in the FreeRADIUS SQL database on the VPS. Each Mikrotik router is configured as a RADIUS client, delegating all authentication and accounting to FreeRADIUS. This architecture provides centralized management, consistent accounting across multiple routers, and instant voucher availability without needing the target router to be online at creation time.

3.4.1 RADIUS Architecture for Vouchers

Each voucher is represented as a set of RADIUS attributes stored across the standard FreeRADIUS SQL schema tables:

- radcheck: Stores authentication credentials (Cleartext-Password or NT-Password) and per-user access control attributes
- radreply: Stores reply attributes sent back to the router on successful authentication (e.g., bandwidth limits, session timeouts)
- radusergroup: Associates each voucher with a RADIUS group profile that defines the service plan
- radgroupcheck / radgroupreply: Define group-level profiles (bandwidth plans) with shared attributes like Mikrotik-Rate-Limit, Session-Timeout, Max-All-Session
- radacct: Stores all accounting data — session start/stop, bytes in/out, session duration — populated automatically by RADIUS accounting packets from the router

Mikrotik routers are configured with: Hotspot server RADIUS authentication enabled, RADIUS server pointing to VPS WireGuard tunnel IP (ports 1812/1813), a unique RADIUS shared secret per router, and Accounting enabled with Interim-Update intervals of 300 seconds.

3.4.2 RADIUS Group Profiles (Service Plans)

Instead of Mikrotik's local hotspot user profiles, the system uses FreeRADIUS group profiles to define service plans. Operators can create and manage these profiles through the app, and they are stored in the radgroupcheck and radgroupreply tables.

Attribute	Table	Example Value	Purpose
Mikrotik-Rate-Limit	radgroupreply	2M/2M (or 5M/5M, 10M/10M)	Bandwidth limit (upload/download)
Session-Timeout	radgroupreply	3600 (seconds)	Max duration per session
Max-All-Session	radgroupcheck	86400 (seconds)	Total cumulative time allowed
Max-Total-Octets	radgroupcheck	1073741824 (bytes = 1GB)	Total data limit (up + down)

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Expiration	radcheck (per-user)	2026-04-28T23:59:59	Voucher validity expiry date
Simultaneous-Use	radcheck (per-user)	1	Max concurrent sessions
Mikrotik-Total-Limit	radgroupreply	1073741824	Alternative total data limit via Mikrotik attribute

Operators can create custom group profiles (e.g., “1-Hour-2Mbps”, “24-Hour-5Mbps-1GB”) that combine bandwidth, time, and data limits. When creating a voucher, the operator selects a group profile rather than configuring individual limits manually.

3.4.3 Voucher Creation

Parameter	Description	Options / Validation
Username	Login credential for the hotspot user	Auto-generated (random 6–8 chars) or custom
Password	Authentication password	Auto-generated (random 6–8 chars) or custom
RADIUS Group Profile	Service plan defining bandwidth, time, and data limits	Dropdown from operator’s configured group profiles
Session Time Limit	Maximum online duration per session (override group default)	Minutes, hours, or days (optional override)
Total Time Limit	Maximum cumulative online time (override group default)	Minutes, hours, or days (optional override)
Data Limit (Upload)	Maximum upload data (override group default)	MB or GB (optional override)
Data Limit (Download)	Maximum download data (override group default)	MB or GB (optional override)
Total Data Limit	Combined upload + download limit (override group default)	MB or GB (optional override)
Validity Period	Expiration date/time for the voucher (Expiration attribute in radcheck)	Hours or days from creation (e.g., 24h, 30d)
Simultaneous-Use	Number of concurrent sessions allowed	1–10 (default: 1)
Router Assignment	Which router(s) this voucher is valid on	Single router, multiple routers, or all routers
Comment	Internal note for the operator	Optional, max 255 characters

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When a voucher is created, the backend inserts records into radcheck (credentials + per-user overrides), radreply (per-user reply attributes if any), and radusergroup (group profile assignment). The voucher is immediately available for authentication on any assigned router — the router does not need to be online at creation time since authentication is handled centrally by FreeRADIUS.

3.4.4 Bulk Voucher Creation

Operators can create up to 100 vouchers in a single batch. All vouchers in a batch share the same RADIUS group profile, validity period, and router assignment. Usernames and passwords are auto-generated. The system validates the batch size against the operator's remaining monthly quota before processing. Bulk inserts are executed as a single database transaction for atomicity.

3.4.5 Voucher Actions

- View: Display voucher details including real-time usage statistics from radacct (time consumed, data consumed, sessions)
- Share: Send voucher credentials via WhatsApp, SMS, or any share-capable app on the device
- Print: Generate a printable voucher card (formatted for thermal receipt printers or A4 paper with multiple vouchers per page)
- Disable: Set Auth-Type := Reject in radcheck to temporarily block authentication without deleting the voucher
- Enable: Remove Auth-Type := Reject to reactivate a previously disabled voucher
- Delete: Remove all records from radcheck, radreply, and radusergroup; optionally send RADIUS CoA Disconnect to terminate any active session
- Extend: Modify Expiration attribute or data/time limits to give additional usage to an existing voucher

3.5 Session Monitoring

Session data is sourced from two places: real-time active sessions are queried from the Mikrotik router via the RouterOS API, and historical session data is stored in the FreeRADIUS radacct table, populated automatically by RADIUS accounting packets.

3.5.1 Active Sessions

Data Point	Description
Username	The hotspot username currently connected
IP Address	Assigned IP address on the hotspot network
MAC Address	Client device's MAC address

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Uptime	Duration of the current session
Data Used (Up/Down)	Bytes transferred in the current session
Idle Time	Time since last data transfer
Login Time	Timestamp when the session started

Active sessions refresh automatically every 30 seconds. Operators can manually force a refresh. Each session has a “Disconnect” button that sends a RADIUS CoA Disconnect-Request (RFC 5176) to the router, terminating the session immediately and triggering a RADIUS Accounting Stop packet.

3.5.2 Session History

Session history is sourced from the FreeRADIUS radacct table, which stores comprehensive accounting data for every session. The system retains session logs for the past 90 days (compared to 30 days for local-only storage). History includes: AcctStartTime, AcctStopTime, AcctSessionTime (duration), AcctInputOctets and AcctOutputOctets (data consumed), AcctTerminateCause (timeout, manual disconnect, data limit, user logout, NAS reboot), FramedIPAddress, CallingStationId (MAC address), and NASIPAddress (router). Operators can search and filter by username, date range, router, or termination cause.

3.6 Dashboard & Reporting

3.6.1 Dashboard Widgets

Widget	Data Shown	Refresh Rate
Active Sessions	Count of currently connected users per router	30 seconds
Vouchers Created Today	Number of vouchers created in the current day	Real-time
Data Usage (24h)	Total upload + download across all sessions in last 24 hours	5 minutes
Peak Hours	Bar chart of session counts by hour over the last 7 days	Hourly
Router Status	Online/offline status indicators for all registered routers	60 seconds
Subscription Status	Current plan, days remaining, quota usage	On page load
Revenue Estimate	Vouchers sold multiplied by operator-set price per voucher	Daily

3.6.2 Reports & Export

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Reports are available for Professional and Enterprise tier subscribers. Reports can be generated for custom date ranges and exported in PDF or CSV format.

- Voucher Sales Report: Vouchers created, used, expired, and remaining by date
- Session Report: Total sessions, average duration, total data consumed
- Revenue Report: Estimated revenue based on voucher pricing set by the operator
- Router Uptime Report: Online/offline history, average uptime percentage

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3.7 Push Notifications

Event	Trigger	Timing
Subscription Expiring	Days remaining reaches threshold	7 days, 3 days, 1 day before
Subscription Expired	Subscription end date reached	Immediately
Payment Confirmed	Admin activates subscription	Immediately
Router Offline	Router status changes to Offline	After 3-minute grace period
Router Back Online	Router status returns to Online	Immediately
Voucher Quota Low	Monthly quota below 10%	Once when threshold crossed
Bulk Creation Complete	Large batch finishes processing	On completion

Users can configure notification preferences in the app settings, enabling or disabling each notification category individually.

3.8 Admin Web Panel

A web-based administration panel accessible to platform operators (not end users). The panel provides centralized control over user accounts, subscriptions, and platform health.

3.8.1 Admin Features

- User Management: View, search, edit, suspend, and delete user accounts
- Subscription Management: Activate, extend, downgrade, or cancel subscriptions manually
- Payment Verification: Queue of pending payment receipts with approve/reject actions
- Platform Statistics: Total users, active subscriptions, total routers, total vouchers created, system health
- Router Overview: List of all registered routers across all users with status indicators
- Audit Logs: Timestamped log of all admin actions

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4. API Specification (Summary)

All API endpoints are prefixed with /api/v1/ and require a valid JWT in the Authorization header (Bearer token) unless marked as public.

4.1 Authentication Endpoints

Method	Endpoint	Description	Auth
POST	/auth/register	Create new user account	Public
POST	/auth/login	Authenticate and receive JWT	Public
POST	/auth/refresh	Refresh JWT using refresh token	Public
POST	/auth/forgot-password	Initiate password reset	Public
POST	/auth/reset-password	Complete password reset with OTP	Public
POST	/auth/logout	Invalidate current session	Required

4.2 Router Endpoints

Method	Endpoint	Description
GET	/routers	List all routers for the authenticated user
POST	/routers	Register a new router, generate WireGuard and RADIUS client config
GET	/routers/:id	Get router details and status
PUT	/routers/:id	Update router name or API credentials
DELETE	/routers/:id	Remove router, destroy WireGuard tunnel, and remove RADIUS NAS entry
GET	/routers/:id/status	Get real-time router status and system info
GET	/routers/:id/setup-guide	Get step-by-step RADIUS and WireGuard setup instructions for this router

4.3 Voucher Endpoints

Method	Endpoint	Description
GET	/routers/:id/vouchers	List all RADIUS vouchers assigned to a router (paginated)
POST	/routers/:id/vouchers	Create a single voucher in RADIUS database

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POST	/routers/:id/vouchers/bulk	Create multiple vouchers in batch (single transaction)
GET	/routers/:id/vouchers/:vid	Get voucher details and usage stats from radacct
PATCH	/routers/:id/vouchers/:vid	Enable, disable, or extend a voucher
DELETE	/routers/:id/vouchers/:vid	Delete voucher from RADIUS and send CoA Disconnect if active

4.4 RADIUS Profile Endpoints

Method	Endpoint	Description
GET	/profiles	List all RADIUS group profiles for the authenticated user
POST	/profiles	Create a new RADIUS group profile (bandwidth, time, data limits)
GET	/profiles/:pid	Get profile details and associated attributes
PUT	/profiles/:pid	Update profile attributes (affects new vouchers only)
DELETE	/profiles/:pid	Delete profile (fails if vouchers are still assigned)

4.5 Session Endpoints

Method	Endpoint	Description
GET	/routers/:id/sessions	List active sessions on a router (via RouterOS API)
DELETE	/routers/:id/sessions/:sid	Disconnect session via RADIUS CoA Disconnect-Request
GET	/routers/:id/sessions/history	Retrieve session history from radacct (paginated, filterable)

4.6 Subscription & Payment Endpoints

Method	Endpoint	Description
GET	/subscription	Get current subscription status and quota
POST	/subscription/request	Submit a subscription or renewal request
POST	/subscription/receipt	Upload payment receipt for admin review
GET	/subscription/plans	List available subscription plans and pricing

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5. Non-Functional Requirements

5.1 Performance

Metric	Target	Measurement Method
Voucher Creation (single)	< 2 seconds end-to-end	From app button press to RADIUS DB insert confirmation
Voucher Creation (bulk, 100)	< 15 seconds end-to-end	From app submission to batch transaction commit
RADIUS Authentication	< 200ms per Access-Request	FreeRADIUS processing time (SQL query + response)
Active Session Refresh	< 2 seconds	API response time for session list via RouterOS API
App Launch to Dashboard	< 3 seconds on 3G connection	Cold start to interactive dashboard
API Response (95th percentile)	< 500ms	Server-side latency measurement
Concurrent RADIUS Auth	200+ simultaneous auth requests	Load testing with simulated NAS clients
Concurrent App Users	500+ simultaneous users	Load testing with simulated traffic

5.2 Security

- TLS 1.3 for all client-server communication; HTTP Strict Transport Security (HSTS) enabled
- WireGuard VPN for all server-to-router communication; no plaintext API traffic
- AES-256 encryption for all sensitive data at rest (API credentials, personal data)
- bcrypt with cost factor 12 for password hashing
- JWT access tokens: 15-minute expiry; refresh tokens: 7-day expiry with rotation
- API rate limiting: 100 requests/minute/user; 10 requests/minute on auth endpoints
- OWASP Top 10 compliance: input validation, parameterized queries, CSRF protection
- Annual third-party penetration testing

5.3 Reliability & Availability

- Backend uptime target: 99.5% (approximately 3.65 hours downtime per month)
- Automated database backups every 6 hours, retained for 30 days
- Disaster recovery: full system restore within 4 hours from backup

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- Graceful degradation: if a router is offline, the app shows cached data and queues commands

5.4 Compatibility

- Android: API level 26+ (Android 8.0 Oreo and above); minimum 2 GB RAM
- iOS: 13.0 and above
- Mikrotik RouterOS: Version 6.48+ and 7.x (both long-term and stable channels)
- Admin Web Panel: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

5.5 Localization

Phase 1 (MVP) will be English only. Phase 2 will add support for French, Portuguese, Swahili, and Arabic to serve the primary target markets in Africa and the Middle East. All user-facing strings will be externalized from code using i18n resource files.

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6. Database Schema (Key Entities)

The system uses a single PostgreSQL database shared between the application backend and FreeRADIUS (via rlm_sql_postgresql). Application-specific tables manage users, subscriptions, and routers, while FreeRADIUS standard schema tables (radcheck, radreply, radusergroup, radgroupcheck, radgroupreply, radacct, nas) handle voucher authentication and accounting.

6.1 Application Tables

Entity	Key Fields	Relationships
Users	id, name, email, phone, password_hash, business_name, is_verified, created_at	Has many Routers, has one Subscription
Subscriptions	id, user_id, plan_tier, start_date, end_date, status, voucher_quota, vouchers_used	Belongs to User
Routers	id, user_id, name, model, ros_version, api_user, api_pass_enc, wg_public_key, wg_private_key_enc, tunnel_ip, radius_secret_enc, nas_identifier, status, last_seen	Belongs to User
Voucher_Meta	id, user_id, router_id, radius_username, group_profile, comment, created_at, status	Belongs to User and Router; links to radcheck via radius_username
RADIUS_Profiles	id, user_id, group_name, display_name, bandwidth_up, bandwidth_down, session_timeout, total_time, total_data, description	Belongs to User; maps to radgroupcheck/radgroupreply
Payments	id, user_id, plan_tier, amount, reference_code, receipt_url, status, reviewed_by, created_at	Belongs to User
Audit Logs	id, admin_id, action, target_entity, target_id, details, timestamp	Belongs to Admin User

6.2 FreeRADIUS Standard Tables

Table	Purpose	Key Fields
radcheck	Per-user authentication attributes	id, username, attribute, op, value (e.g., Cleartext-Password, Expiration)
radreply	Per-user reply attributes sent on Access-Accept	id, username, attribute, op, value (e.g., per-user overrides)
radusergroup	Maps users to group profiles	id, username, groupname, priority

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radgroupcheck	Group-level check attributes (limits)	id, groupname, attribute, op, value (e.g., Max-All-Session, Max-Total-Octets)
radgroupreply	Group-level reply attributes (bandwidth)	id, groupname, attribute, op, value (e.g., Mikrotik-Rate-Limit, Session-Timeout)
radacct	Session accounting records	radacctid, username, nasipaddress, acctstarttime, acctstoptime, acctsessiontime, acctinputoctets, acctoutputoctets, acctterminatecause, framedipaddress, callingstationid
nas	Registered RADIUS clients (routers)	id, nasname, shortname, type, secret, description

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7. UI/UX Guidelines

7.1 Design Principles

- Speed first: The app must feel fast. Skeleton screens while loading, optimistic UI updates for voucher creation.
- Clarity over decoration: Every screen should have a single primary action. Avoid visual clutter.
- Offline tolerance: Show cached data when offline, queue actions, and sync when connectivity returns.
- Accessible: Minimum touch target 44x44pt, sufficient color contrast (WCAG AA), support for system font scaling.

7.2 Key Screens

Screen	Purpose	Primary Action
Login / Register	Authentication entry point	Sign In / Create Account
Dashboard	Overview of all routers and key metrics	Quick-create voucher
Router List	List of registered routers with status	Add Router
Router Detail	Single router info, status, system stats	Create Voucher / View Sessions
RADIUS Profiles	Manage group profiles (bandwidth plans)	Create Profile
Voucher Creation	Configure and create new voucher(s)	Create (single) / Generate (bulk)
Voucher List	All vouchers on a router with filters	Share / Print / Delete
Active Sessions	Currently connected users on a router	Disconnect
Session History	Past session logs with search and filter	Export
Subscription	Current plan, quota usage, renew option	Renew / Upgrade
Settings	Profile, notifications, app preferences	Save Changes

7.3 Navigation

The app uses a bottom tab bar with four primary tabs: Dashboard, Routers, Vouchers, and Settings. The Vouchers tab shows vouchers across all routers with a router filter. RADIUS group profiles are managed from a dedicated section within the Routers or Settings tab. Session monitoring is accessed from within a router’s detail screen.

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8. Release Plan

8.1 Phase 1 — MVP

Target: 8 weeks from development start

Goal: Core functionality sufficient for a paying customer to manage vouchers remotely.

Feature	Scope	Priority
User Registration & Login	Email/password auth, JWT sessions, email verification	P0 — Must Have
Manual Subscription	Starter tier only, bank transfer, admin activation	P0 — Must Have
Router Connection	Add one router via WireGuard + RADIUS client config, online/offline status	P0 — Must Have
RADIUS Profiles	Create and manage group profiles (bandwidth, time, data limits)	P0 — Must Have
Voucher Creation	Single and bulk creation in RADIUS database with group profile assignment	P0 — Must Have
Active Session View	List active sessions with disconnect capability	P0 — Must Have
Basic Dashboard	Active session count, vouchers created today, router status	P1 — Should Have
Share Voucher	Share credentials via device share sheet	P1 — Should Have

8.2 Phase 2 — Growth

Target: 6 weeks after Phase 1 launch

Feature	Scope	Priority
Advanced Reports	Custom date ranges, PDF/CSV export	P1 — Should Have
Bulk Voucher Printing	Thermal printer format, A4 multi-voucher layout	P1 — Should Have
Push Notifications	Subscription, router, and quota alerts via FCM/APNs	P1 — Should Have
Multi-language Support	French, Portuguese, Swahili, Arabic	P2 — Nice to Have

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Admin Web Panel	User and subscription management, payment queue	P1 — Should Have
Professional & Enterprise Tiers	Multi-router support, higher quotas	P1 — Should Have
Session History	30-day log with search, filter, and export	P1 — Should Have
Biometric Login	Fingerprint / Face ID convenience login	P2 — Nice to Have

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9. Out of Scope

The following items are explicitly excluded from this product version and will not be developed unless a future PRD revision adds them:

- Automated payment processing (credit card, mobile money integration, payment gateways)
- Support for non-Mikrotik routers (Ubiquiti, Cisco, TP-Link, etc.)
- End-user self-registration app (a customer-facing app where end users buy their own vouchers)
- Bandwidth throttling or QoS management beyond what RADIUS group profiles and Mikrotik-Rate-Limit attributes provide
- White-label or reseller branding of the application
- Integration with third-party accounting or invoicing software
- SMS-based voucher delivery (would require SMS gateway integration)

10. Success Metrics & KPIs

Metric	Target	Measurement Period
Active Paying Subscribers	100	Within 3 months of launch
Voucher Creation Time	Under 1 minute (end-to-end)	Ongoing
Reduction in Technician Visits	90%	Surveyed at 3 months post-launch
App Store Rating	4.0+ stars	Ongoing
Monthly Churn Rate	< 5%	Monthly after Month 2
Backend Uptime	99.5%	Monthly
Customer Support Response	< 24 hours	Ongoing
Voucher Success Rate	99%+ (created without error)	Ongoing

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11. Risks & Mitigations

Risk	Impact	Likelihood	Mitigation
RouterOS API changes in future versions	Medium	Medium	RouterOS API used only for monitoring and config; voucher logic is RADIUS-based and router-independent
FreeRADIUS database corruption or failure	Critical	Low	Automated PostgreSQL backups every 6 hours; database replication; tested restore procedures
RADIUS authentication latency over WireGuard	High	Medium	Optimize FreeRADIUS SQL queries; use connection pooling; monitor round-trip times; local DNS caching on routers
WireGuard tunnel instability on poor networks	High	Medium	Persistent keepalive, exponential backoff reconnection, Mikrotik fallback to local users if RADIUS unreachable
Manual payment verification bottleneck	Medium	High	Streamlined admin queue UI; target < 2 hour verification turnaround; Phase 3 automation
Low initial adoption	High	Medium	Target existing Mikrotik communities; offer free 7-day trial; partner with local ISPs
Security breach or data leak	Critical	Low	Encryption at rest and in transit; penetration testing; minimal data retention; incident response plan
App store rejection	Medium	Low	Follow platform guidelines strictly; no side-loading requirements; clear privacy policy
Router misconfiguration of RADIUS settings	Medium	High	In-app step-by-step setup wizard; automated config validation via RouterOS API; troubleshooting guide

12. Appendices

12.1 Glossary

Term	Definition
RouterOS	Mikrotik’s proprietary operating system for their router hardware
Hotspot	A Mikrotik feature that provides captive portal authentication for Wi-Fi users

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Voucher	A set of login credentials (username + password) stored in the RADIUS database for hotspot authentication
FreeRADIUS	Open-source RADIUS server implementing the AAA (Authentication, Authorization, Accounting) protocol; used as the centralized voucher authentication engine
RADIUS	Remote Authentication Dial-In User Service — a networking protocol for centralized authentication and accounting (RFC 2865/2866)
radcheck	FreeRADIUS SQL table storing per-user authentication attributes (credentials, expiration, limits)
radacct	FreeRADIUS SQL table storing session accounting records (start/stop times, data usage, duration)
NAS	Network Access Server — in this context, the Mikrotik router acting as a RADIUS client
CoA	Change of Authorization (RFC 5176) — RADIUS mechanism to dynamically modify or disconnect active sessions
Mikrotik-Rate-Limit	Vendor-specific RADIUS attribute used to set bandwidth limits on Mikrotik hotspot sessions
rlm_sql	FreeRADIUS module that enables SQL database backend for user storage and accounting
WireGuard	A modern, high-performance VPN protocol built into the Linux kernel
JWT	JSON Web Token — a compact, URL-safe token format for authentication claims
VPS	Virtual Private Server — a cloud-hosted virtual machine
OTP	One-Time Password — a temporary code for verification

12.2 Revision History

Version	Date	Author	Changes
1.0	2026-03-28	Product Team	Initial draft — complete PRD

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